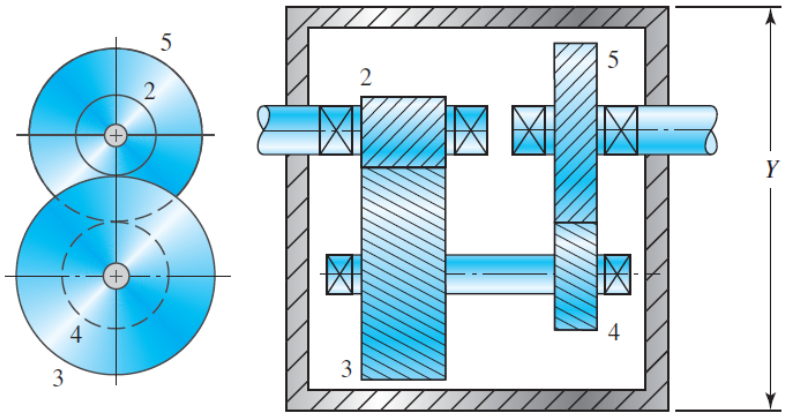
KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF MECHANICAL ENGINEERING

EQUATIONS, TABLES AND CHARTS

ME 374: MACHINE ELEMENT DESIGN II



Compiled By

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Reference Textbook:

Shigley's Mechanical Engineering Design, 10th edition

By Richard Budynas, and J. Nisbett, McGraw Hill

DESIGN OF SPUR GEARS: EQUATIONS, TABLES AND CHARTS

EQUATIONS

$$P = \frac{N}{d} \qquad (13\text{–}1)$$

$$m = \frac{d}{N} \qquad (13\text{–}2)$$

$$W_t = \frac{60000H}{\pi dn} \qquad (13\text{-}36)$$

Lewis Equation $$\sigma = \frac{K_v W^t}{FmY} \qquad (14\text{–}8)$$

$$K_V = \frac{3.05 + V}{3.05} \quad \text{(cast iron, cast profile)} \qquad V \text{ in m/s}$$

$$K_V = \frac{6.1 + V}{6.1} \quad \text{(cut or milled profile)} \qquad V \text{ in m/s}$$

$$K_V = \frac{3.56 + \sqrt{V}}{3.56} \quad \text{(hobbed or shaped profile)} \qquad V \text{ in m/s}$$

$$K_V = \sqrt{\frac{5.56 + \sqrt{V}}{5.56}} \quad \text{(shaved or ground profile)} \qquad V \text{ in m/s}$$

AGMA Bending Stress, $$\sigma = \begin{cases} W^t K_o K_V K_S \dfrac{P_d}{F} \dfrac{K_m K_B}{J} & \text{US Customary Units} \\ W^t K_o K_V K_S \dfrac{1}{bm_t} \dfrac{K_H K_B}{Y_J} & \text{S.I. Units} \end{cases}$$

AGMA Allowable Bending Stress,

$$\sigma_{all} = \begin{cases} \dfrac{S_t}{S_F} \dfrac{Y_N}{K_T K_R} & \text{(U.S. customary units)} \\ \dfrac{S_t}{S_F} \dfrac{Y_N}{Y_\theta Y_Z} & \text{(SI units)} \end{cases} \qquad (14\text{–}17)$$

AGMA Contact Stress, $$\sigma_C = \begin{cases} C_p \sqrt{W^t K_O K_V K_S \dfrac{K_m C_f}{d_p F I}} & \text{US Customary Units} \\ Z_E \sqrt{W^t K_O K_V K_S \dfrac{K_H Z_R}{d_{w1} b Z_I}} & \text{S.I. Units} \end{cases}$$

AGMA Allowable Contact Stress,

$$\sigma_{c,\text{all}} = \begin{cases} \dfrac{S_c}{S_H}\dfrac{Z_N C_H}{K_T K_R} & \text{(U.S. customary units)} \\ \dfrac{S_c}{S_H}\dfrac{Z_N Z_W}{Y_\theta Y_Z} & \text{(SI units)} \end{cases} \qquad (14\text{–}18)$$

$$C_p = \left[\frac{1}{\pi \left(\dfrac{1-\nu_P^2}{E_P} + \dfrac{1-\nu_G^2}{E_G} \right)} \right]^{1/2} \qquad (14\text{–}13)$$

$$r_1 = \frac{d_p \sin\phi}{2} \qquad r_2 = \frac{d_G \sin\phi}{2} \qquad (14\text{-}12)$$

$$\sigma_C = -C_p \left[\frac{K_V W^t}{F \cos\phi} \left(\frac{1}{r_1} + \frac{1}{r_2} \right) \right]^{\frac{1}{2}} \qquad (14\text{-}14)$$

$$m_G = \frac{N_G}{N_P} = \frac{d_G}{d_P} \qquad (14\text{–}22)$$

$$m_N = \frac{p_N}{0.95Z} \qquad (14\text{–}21)$$

$$p_N = p_n \cos\phi_n \qquad (14\text{–}24)$$

$$Z = \left[(r_P + a)^2 - r_{bP}^2 \right]^{1/2} + \left[(r_G + a)^2 - r_{bG}^2 \right]^{1/2} - (r_P + r_G)\sin\phi_t \qquad (14\text{–}25)$$

$$r_b = r\cos\phi_t \qquad (14\text{–}26)$$

$$I = \begin{cases} \dfrac{\cos\phi_t \sin\phi_t}{2m_N} \dfrac{m_G}{m_G + 1} & \text{external gears} \\ \dfrac{\cos\phi_t \sin\phi_t}{2m_N} \dfrac{m_G}{m_G - 1} & \text{internal gears} \end{cases} \qquad (14\text{-}23)$$

$$K_V = \begin{cases} \left(\dfrac{A + \sqrt{V}}{A} \right)^B & \text{V in ft/min} \\ \left(\dfrac{A + \sqrt{200V}}{A} \right)^B & \text{V in m/s} \end{cases} \qquad (14\text{-}27)$$

$$A = 50 + 56(1 - B)$$
$$B = 0.25(12 - Q_V)^{2/3} \qquad (14\text{-}28)$$

$$K_S = \frac{1}{K_b} = 1.192\left(\frac{F\sqrt{Y}}{P}\right)^{0.0535}$$

$$K_R = \begin{cases} 0.658 - 0.0759\text{In}(1-\text{R}) & 0.5 < \text{R} < 0.99 \\ 0.50 - 0.109\text{In}(1-\text{R}) & 0.99 \le \text{R} \le 0.9999 \end{cases} \quad (14\text{-}38)$$

$$S_F = \frac{S_t Y_N / (K_T K_R)}{\sigma} \quad (14\text{-}41)$$

$$S_H = \frac{S_C Z_N C_H / (K_T K_R)}{\sigma_C} \quad (14\text{-}42)$$

TABLES

Table 14–2

Values of the Lewis Form Factor Y (These Values Are for a Normal Pressure Angle of 20°, Full-Depth Teeth, and a Diametral Pitch of Unity in the Plane of Rotation)

Number of Teeth	Y	Number of Teeth	Y
12	0.245	28	0.353
13	0.261	30	0.359
14	0.277	34	0.371
15	0.290	38	0.384
16	0.296	43	0.397
17	0.303	50	0.409
18	0.309	60	0.422
19	0.314	75	0.435
20	0.322	100	0.447
21	0.328	150	0.460
22	0.331	300	0.472
24	0.337	400	0.480
26	0.346	Rack	0.485

Table of Overload Factors, K_o

	Driven Machine		
Power source	Uniform	Moderate shock	Heavy shock
Uniform	1.00	1.25	1.75
Light shock	1.25	1.50	2.00
Medium shock	1.50	1.75	2.25

TABLE 15.2 Mounting Correction Factor K_m

Characteristics of Support	Face Width (in.) 0 to 2	6	9	16 up
Accurate mountings, small bearing clearances, minimum deflection, precision gears	1.3	1.4	1.5	1.8
Less rigid mountings, less accurate gears, contact across the full face	1.6	1.7	1.8	2.2
Accuracy and mounting such that less than full-face contact exists		Over 2.2		

Reliability	K_R (Y_Z)
0.9999	1.50
0.999	1.25
0.99	1.00
0.90	0.85
0.50	0.70

Table 14–3

Repeatedly Applied Bending Strength S_t at 10^7 Cycles and 0.99 Reliability for Steel Gears

Source: ANSI/AGMA 2001-D04.

Material Designation	Heat Treatment	Minimum Surface Hardness[1]	Allowable Bending Stress Number S_t,[2] psi Grade 1	Grade 2	Grade 3
Steel[3]	Through-hardened	See Fig. 14–2 See Table 8*	See Fig. 14–2 45 000	See Fig. 14–2 55 000	— —
	Flame[4] or induction hardened[4] with type A pattern[5]	See Table 8*	22 000	22 000	—
	Flame[4] or induction hardened[4] with type B pattern[5]	See Table 9*	55 000	65 000 or 70 000[6]	75 000
	Carburized and hardened	83.5 HR15N	See Fig. 14–3	See Fig. 14–3	—
	Nitrided[4,7] (through-hardened steels)	87.5 HR15N	See Fig. 14–4	See Fig. 14–4	See Fig. 14–4
Nitralloy 135M, Nitralloy N, and 2.5% chrome (no aluminum)	Nitrided[4,7]				

Table 14–8

Elastic Coefficient C_p (Z_E), $\sqrt{\text{psi}}$ ($\sqrt{\text{MPa}}$) *Source: AGMA 218.01*

Pinion Material	Pinion Modulus of Elasticity E_P psi (MPa)*	Gear Material and Modulus of Elasticity E_G, lbf/in² (MPa)* Steel 30×10^6 (2×10^5)	Malleable Iron 25×10^6 (1.7×10^5)	Nodular Iron 24×10^6 (1.7×10^5)	Cast Iron 22×10^6 (1.5×10^5)	Aluminum Bronze 17.5×10^6 (1.2×10^5)	Tin Bronze 16×10^6 (1.1×10^5)
Steel	30×10^6 (2×10^5)	2300 (191)	2180 (181)	2160 (179)	2100 (174)	1950 (162)	1900 (158)
Malleable iron	25×10^6 (1.7×10^5)	2180 (181)	2090 (174)	2070 (172)	2020 (168)	1900 (158)	1850 (154)
Nodular iron	24×10^6 (1.7×10^5)	2160 (179)	2070 (172)	2050 (170)	2000 (166)	1880 (156)	1830 (152)
Cast iron	22×10^6 (1.5×10^5)	2100 (174)	2020 (168)	2000 (166)	1960 (163)	1850 (154)	1800 (149)
Aluminum bronze	17.5×10^6 (1.2×10^5)	1950 (162)	1900 (158)	1880 (156)	1850 (154)	1750 (145)	1700 (141)
Tin bronze	16×10^6 (1.1×10^5)	1900 (158)	1850 (154)	1830 (152)	1800 (149)	1700 (141)	1650 (137)

Poisson's ratio = 0.30.

*When more exact values for modulus of elasticity are obtained from roller contact tests, they may be used.

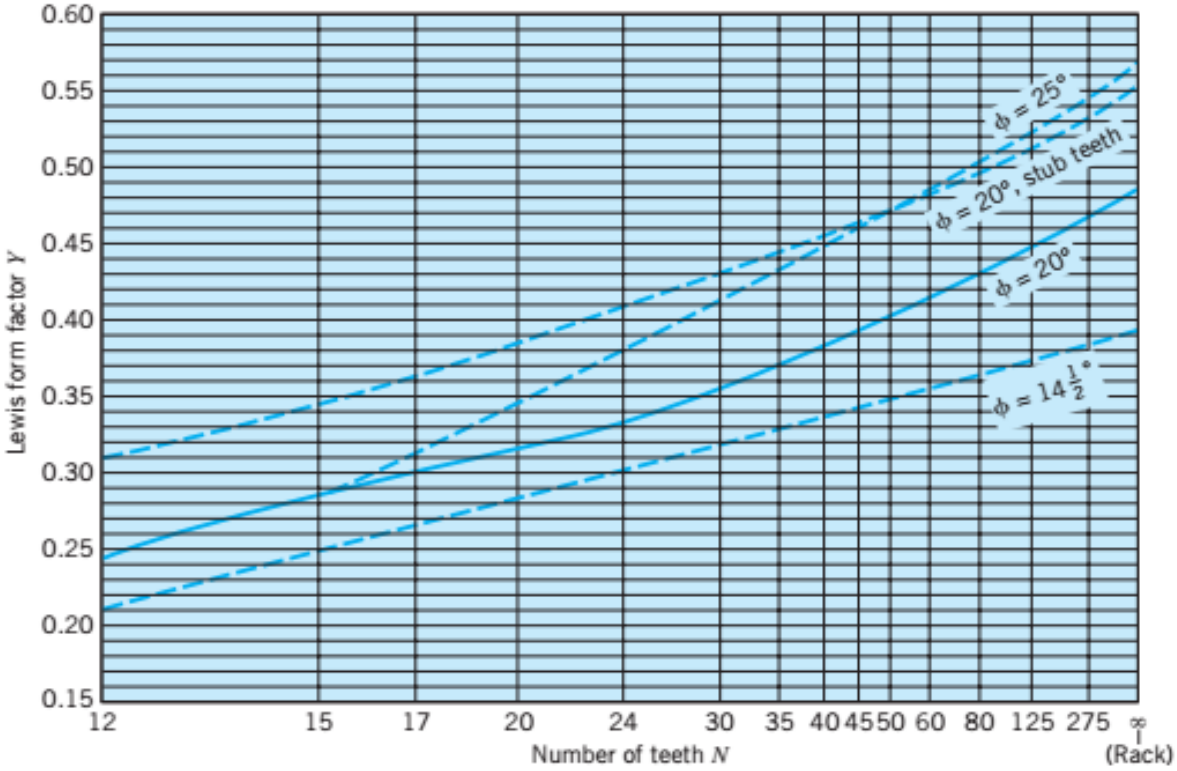
Table 14–6

Repeatedly Applied Contact Strength S_c at 10^7 Cycles and 0.99 Reliability for Steel Gears

Source: ANSI/AGMA 2001-D04.

Material Designation	Heat Treatment	Minimum Surface Hardness[1]	Allowable Contact Stress Number,[2] S_c, psi Grade 1	Grade 2	Grade 3
Steel[3]	Through hardened[4]	See Fig. 14–5	See Fig. 14–5	See Fig. 14–5	—
	Flame[5] or induction hardened[5]	50 HRC	170 000	190 000	—
		54 HRC	175 000	195 000	—
	Carburized and hardened[5]	See Table 9*	180 000	225 000	275 000
	Nitrided[5] (through hardened steels)	83.5 HR15N	150 000	163 000	175 000
		84.5 HR15N	155 000	168 000	180 000
2.5% chrome (no aluminum)	Nitrided[5]	87.5 HR15N	155 000	172 000	189 000
Nitralloy 135M	Nitrided[5]	90.0 HR15N	170 000	183 000	195 000
Nitralloy N	Nitrided[5]	90.0 HR15N	172 000	188 000	205 000
2.5% chrome (no aluminum)	Nitrided[5]	90.0 HR15N	176 000	196 000	216 000

FIGURES/Charts



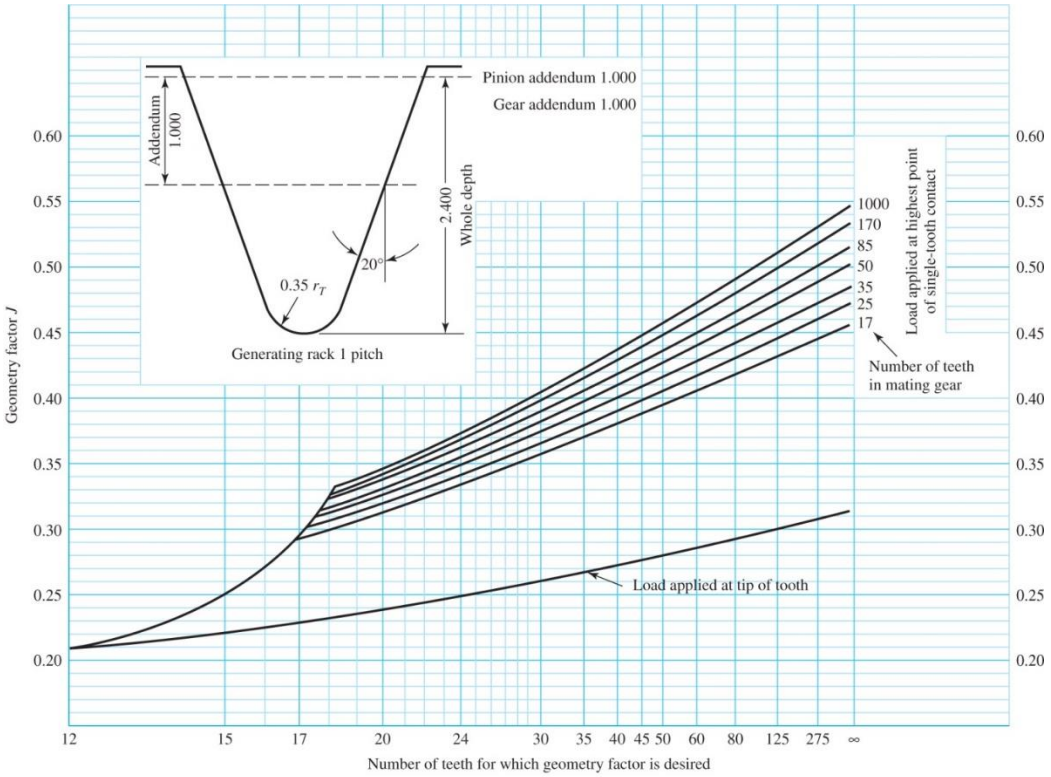


Figure 14–6

Spur-gear geometry factors *J*. *Source:* The graph is from AGMA 218.01, which is consistent with tabular data from the current AGMA 908-B89. The graph is convenient for design purposes.

Figure 14–9

Dynamic factor K_v. The equations to these curves are given by Eq. (14–27) and the end points by Eq. (14–29). *(ANSI/AGMA 2001-D04, Annex A)*

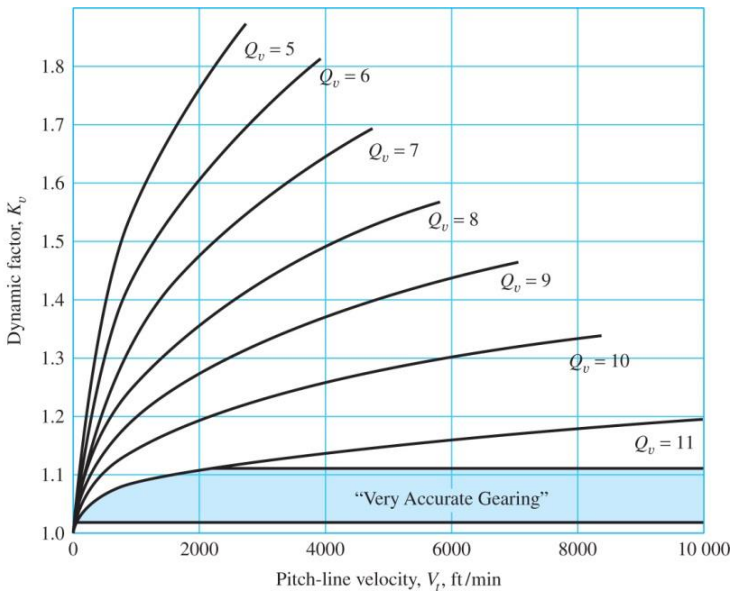


Figure 14–2

Allowable bending stress number for through-hardened steels, S_t. The SI equations are: $S_t = 0.533H_B + 88.3$ MPa, grade 1, and $S_t = 0.703H_B + 113$ MPa, grade 2. *(Source: ANSI/AGMA 2001-D04 and 2101-D04.)*

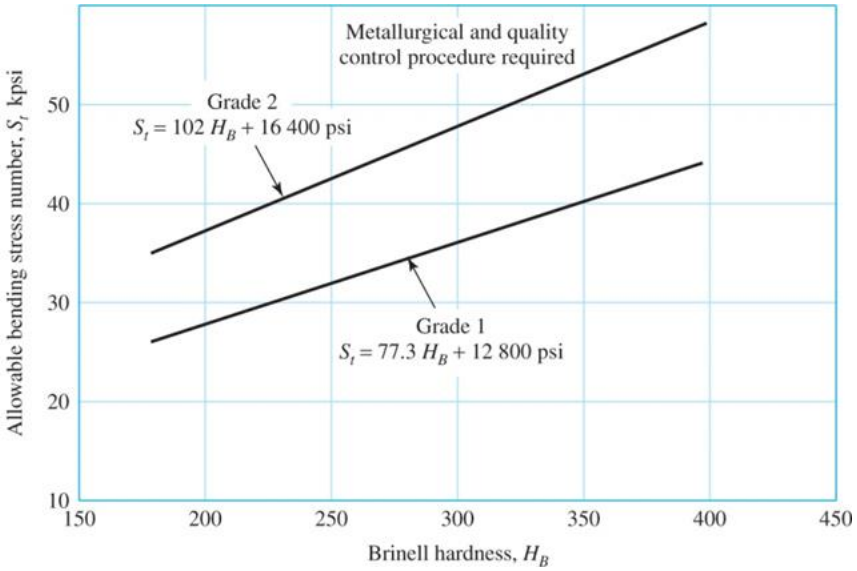


Figure 14–5

Contact-fatigue strength S_c at 10^7 cycles and 0.99 reliability for through-hardened steel gears. The SI equations are: $S_c = 2.22H_B + 200$ MPa, grade 1, and $S_c = 2.41H_B + 237$ MPa, grade 2. *(Source: ANSI/AGMA 2001-D04 and 2101-D04.)*

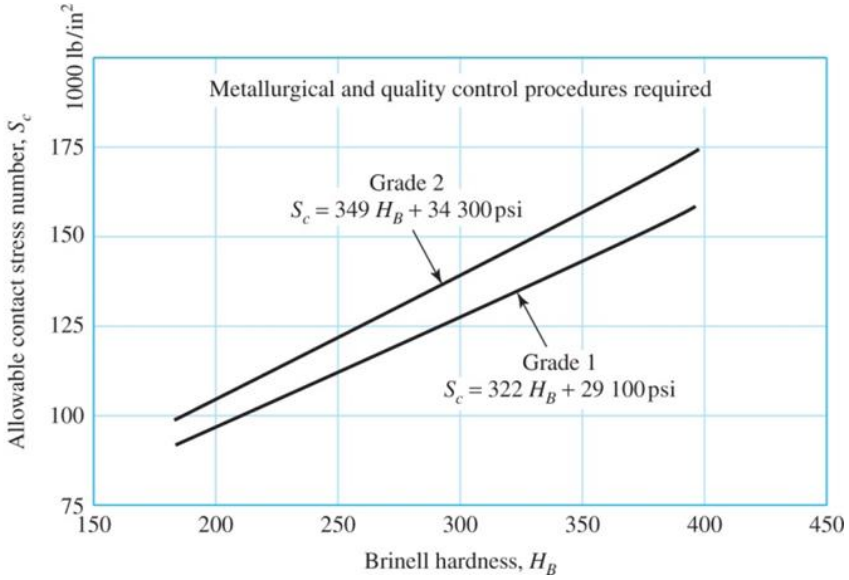


Figure 14–12

Hardness-ratio factor C_H (through-hardened steel). *(ANSI/AGMA 2001-D04.)*

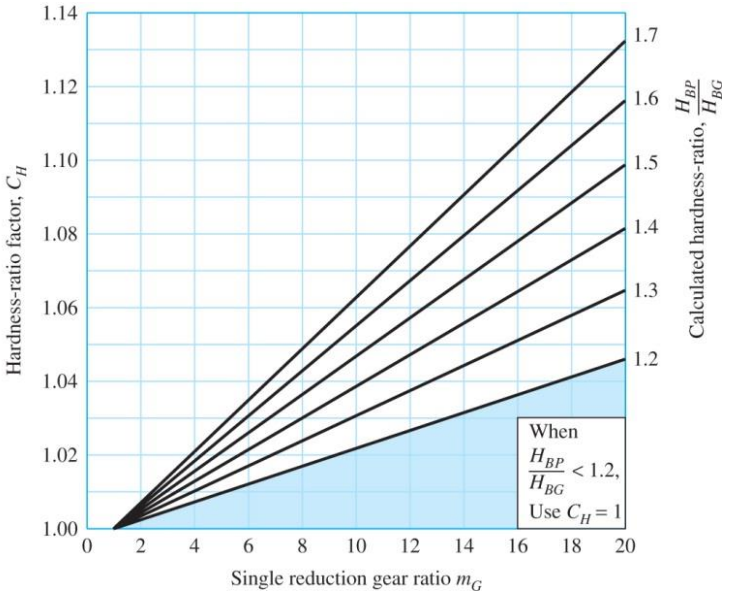


Figure 14–14

Repeatedly applied bending strength stress-cycle factor Y_N. *(ANSI/AGMA 2001-D04.)*

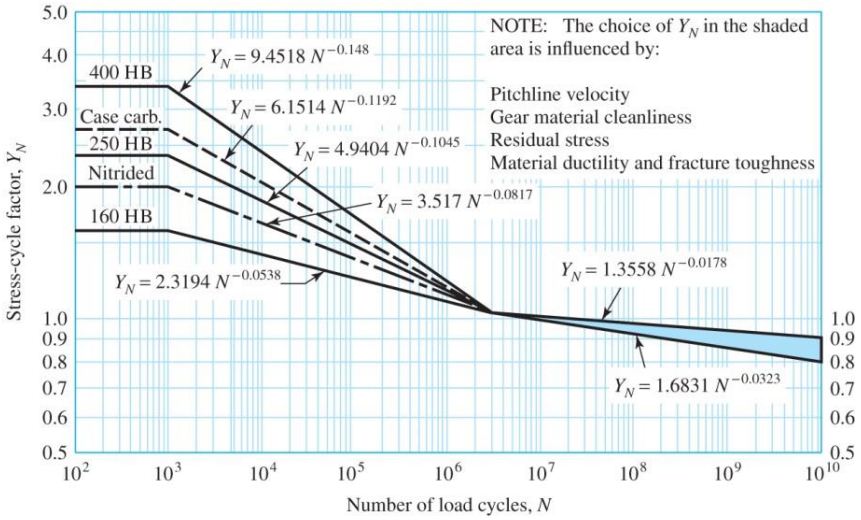
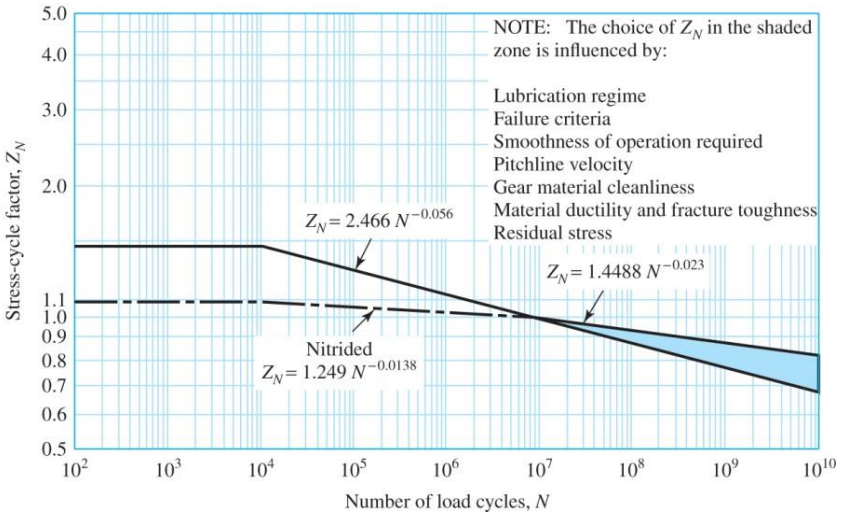


Figure 14–15

Pitting resistance stress-cycle factor Z_N. *(ANSI/AGMA 2001-D04.)*



BELTS

$$\frac{P_1}{P_2} = e^{f\phi} \qquad (16\text{-}19)$$

$$T = (P_1 - P_2)\frac{D}{2} \qquad (16\text{-}20)$$

Vee Belt

$$\frac{P_1 - P_C}{P_2 - P_C} = e^{f\phi/\sin B}$$

Belt Section	Sheave Pitch Diameter, in	Belt Speed, ft/min 1000	2000	3000	4000	5000
A	2.6	0.47	0.62	0.53	0.15	
	3.0	0.66	1.01	1.12	0.93	0.38
	3.4	0.81	1.31	1.57	1.53	1.12
	3.8	0.93	1.55	1.92	2.00	1.71
	4.2	1.03	1.74	2.20	2.38	2.19
	4.6	1.11	1.89	2.44	2.69	2.58
	5.0 and up	1.17	2.03	2.64	2.96	2.89
B	4.2	1.07	1.58	1.68	1.26	0.22
	4.6	1.27	1.99	2.29	2.08	1.24
	5.0	1.44	2.33	2.80	2.76	2.10
	5.4	1.59	2.62	3.24	3.34	2.82
	5.8	1.72	2.87	3.61	3.85	3.45
	6.2	1.82	3.09	3.94	4.28	4.00
	6.6	1.92	3.29	4.23	4.67	4.48
	7.0 and up	2.01	3.46	4.49	5.01	4.90
C	6.0	1.84	2.66	2.72	1.87	
	7.0	2.48	3.94	4.64	4.44	3.12
	8.0	2.96	4.90	6.09	6.36	5.52
	9.0	3.34	5.65	7.21	7.86	7.39
	10.0	3.64	6.25	8.11	9.06	8.89
	11.0	3.88	6.74	8.84	10.0	10.1
	12.0 and up	4.09	7.15	9.46	10.9	11.1
D	10.0	4.14	6.13	6.55	5.09	1.35
	11.0	5.00	7.83	9.11	8.50	5.62
	12.0	5.71	9.26	11.2	11.4	9.18
	13.0	6.31	10.5	13.0	13.8	12.2
	14.0	6.82	11.5	14.6	15.8	14.8
	15.0	7.27	12.4	15.9	17.6	17.0
	16.0	7.66	13.2	17.1	19.2	19.0
	17.0 and up	8.01	13.9	18.1	20.6	20.7
E	16.0	8.68	14.0	17.5	18.1	15.3
	18.0	9.92	16.7	21.2	23.0	21.5
	20.0	10.9	18.7	24.2	26.9	26.4
	22.0	11.7	20.3	26.6	30.2	30.5
	24.0	12.4	21.6	28.6	32.9	33.8
	26.0	13.0	22.8	30.3	35.1	36.7
	28.0 and up	13.4	23.7	31.8	37.1	39.1

CLUTCHES

EQUATIONS

Axial Plates/Uniform Wear

$$F = \int_{\frac{d}{2}}^{\frac{D}{2}} 2\pi p r dr = \pi p_a d \int_{\frac{d}{2}}^{\frac{D}{2}} dr = \frac{\pi p_a d}{2}\left(D - d\right) \qquad (16\text{-}23)$$

$$T = \int_{\frac{d}{2}}^{\frac{D}{2}} 2\pi f p r^2 dr = \pi f p_a d \int_{\frac{d}{2}}^{\frac{D}{2}} r dr = \frac{\pi f p_a d}{8}\left(D^2 - d^2\right) \qquad (16\text{-}24)$$

$$T = \frac{Ff}{4}\left(D + d\right) \qquad (16\text{-}25)$$

Axial Plates/Uniform Pressure

$$F = \frac{\pi p_a}{4}\left(D^2 - d^2\right) \qquad (16\text{-}26)$$

$$T = 2\pi f p_a \int_{\frac{d}{2}}^{\frac{D}{2}} r^2 dr = \frac{\pi f p_a}{12}\left(D^3 - d^3\right) \qquad (16\text{-}27)$$

$$T = \frac{Ff}{3}\frac{D^3 - d^3}{D^2 - d^2} \qquad (16\text{-}28)$$

Cone Clutches/Uniform Wear

$$F = \int pdA\sin\alpha = \int_{\frac{d}{2}}^{\frac{D}{2}} \left(p_a \frac{d}{2r} \right) \left(\frac{2\pi r dr}{\sin\alpha} \right) (\sin\alpha) = \pi p_a \int_{\frac{d}{2}}^{\frac{D}{2}} dr = \frac{\pi p_a d}{2}(D-d) \quad (16\text{-}44)$$

$$T = \int rfpdA = \int_{\frac{d}{2}}^{\frac{D}{2}} (rf) \left(p_a \frac{d}{2r} \right) \left(\frac{2\pi r dr}{\sin\alpha} \right) = \frac{\pi f p_a d}{\sin\alpha} \int_{\frac{d}{2}}^{\frac{D}{2}} r dr = \frac{\pi f p_a d}{8\sin\alpha}(D^2 - d^2) \quad (16\text{-}45)$$

$$T = \frac{Ff}{4\sin\alpha}(D+d) \quad (16\text{-}46)$$

Cone Clutches/Uniform Pressure

$$F = \int p_a dA\sin\alpha = \int_{\frac{d}{2}}^{\frac{D}{2}} (p_a) \left(\frac{2\pi r dr}{\sin\alpha} \right) (\sin\alpha) = \frac{\pi p_a}{4}(D^2 - d^2) \quad (16\text{-}47)$$

$$T = \int rfp_a dA = \int_{\frac{d}{2}}^{\frac{D}{2}} (rfp_a) \left(\frac{2\pi r dr}{\sin\alpha} \right) = \frac{\pi f p_a}{12\sin\alpha}(D^3 - d^3) \quad (16\text{-}48)$$

$$T = \frac{Ff}{3\sin\alpha} \frac{D^3 - d^3}{D^2 - d^2} \quad (16\text{-}49)$$

CHAINS

EQUATIONS

$$D = \frac{p}{\sin(180°/N)} \quad (17\text{-}29)$$

$$V = \frac{Npn}{12} \quad (17\text{-}30)$$

$$H_1 = 0.004 N_1^{1.08} n_1^{0.9} p^{(3-0.07p)} \quad \text{hp} \quad (17\text{-}32)$$

$$H_2 = \frac{1000 K_r N_1^{1.5} p^{0.8}}{n_1^{1.5}} \quad \text{hp} \quad (17\text{-}33)$$

$$\frac{L}{p} \approx \frac{2C}{p} + \frac{N_1 + N_2}{2} + \frac{(N_2 - N_1)^2}{4\pi^2 C/p} \quad (17\text{-}34)$$

$$C = \frac{p}{4}\left[-A + \sqrt{A^2 - 8\left(\frac{N_2 - N_1}{2\pi} \right)^2} \right] \quad (17\text{-}35)$$

$$A = \frac{N_1 + N_2}{2} - \frac{L}{p} \quad (17\text{-}36)$$

$$H_a = K_1 K_2 H_{tab} \qquad (17\text{-}37)$$

$$H_d = H_{\text{nom}} K_s n_d \qquad (17\text{–}38)$$

Chains/TABLES

ANSI Chain Number	Pitch, in (mm)	Width, in (mm)	Minimum Tensile Strength, lbf (N)	Average Weight, lbf/ft (N/m)	Roller Diameter, in (mm)	Multiple-Strand Spacing, in (mm)
25	0.250 (6.35)	0.125 (3.18)	780 (3 470)	0.09 (1.31)	0.130 (3.30)	0.252 (6.40)
35	0.375 (9.52)	0.188 (4.76)	1 760 (7 830)	0.21 (3.06)	0.200 (5.08)	0.399 (10.13)
41	0.500 (12.70)	0.25 (6.35)	1 500 (6 670)	0.25 (3.65)	0.306 (7.77)	— —
40	0.500 (12.70)	0.312 (7.94)	3 130 (13 920)	0.42 (6.13)	0.312 (7.92)	0.566 (14.38)
50	0.625 (15.88)	0.375 (9.52)	4 880 (21 700)	0.69 (10.1)	0.400 (10.16)	0.713 (18.11)
60	0.750 (19.05)	0.500 (12.7)	7 030 (31 300)	1.00 (14.6)	0.469 (11.91)	0.897 (22.78)
80	1.000 (25.40)	0.625 (15.88)	12 500 (55 600)	1.71 (25.0)	0.625 (15.87)	1.153 (29.29)
100	1.250 (31.75)	0.750 (19.05)	19 500 (86 700)	2.58 (37.7)	0.750 (19.05)	1.409 (35.76)
120	1.500 (38.10)	1.000 (25.40)	28 000 (124 500)	3.87 (56.5)	0.875 (22.22)	1.789 (45.44)
140	1.750 (44.45)	1.000 (25.40)	38 000 (169 000)	4.95 (72.2)	1.000 (25.40)	1.924 (48.87)
160	2.000 (50.80)	1.250 (31.75)	50 000 (222 000)	6.61 (96.5)	1.125 (28.57)	2.305 (58.55)
180	2.250 (57.15)	1.406 (35.71)	63 000 (280 000)	9.06 (132.2)	1.406 (35.71)	2.592 (65.84)
200	2.500 (63.50)	1.500 (38.10)	78 000 (347 000)	10.96 (159.9)	1.562 (39.67)	2.817 (71.55)
240	3.00 (76.70)	1.875 (47.63)	112 000 (498 000)	16.4 (239)	1.875 (47.62)	3.458 (87.83)

Table 17–20

Rated Horsepower Capacity of Single-Strand Single-Pitch Roller Chain for a 17-Tooth Sprocket

Source: Compiled from ANSI B29.1-1975 information only section, and from B29.9-1958.

Sprocket Speed, rev/min	ANSI Chain Number					
	25	35	40	41	50	60
50	0.05	0.16	0.37	0.20	0.72	1.24
100	0.09	0.29	0.69	0.38	1.34	2.31
150	0.13*	0.41*	0.99*	0.55*	1.92*	3.32
200	0.16*	0.54*	1.29	0.71	2.50	4.30
300	0.23	0.78	1.85	1.02	3.61	6.20
400	0.30*	1.01*	2.40	1.32	4.67	8.03
500	0.37	1.24	2.93	1.61	5.71	9.81
600	0.44*	1.46*	3.45*	1.90*	6.72*	11.6
700	0.50	1.68	3.97	2.18	7.73	13.3
800	0.56*	1.89*	4.48*	2.46*	8.71*	15.0
900	0.62	2.10	4.98	2.74	9.69	16.7
1000	0.68*	2.31*	5.48	3.01	10.7	18.3
1200	0.81	2.73	6.45	3.29	12.6	21.6
1400	0.93*	3.13*	7.41	2.61	14.4	18.1
1600	1.05*	3.53*	8.36	2.14	12.8	14.8
1800	1.16	3.93	8.96	1.79	10.7	12.4
2000	1.27*	4.32*	7.72*	1.52*	9.23*	10.6
2500	1.56	5.28	5.51*	1.10*	6.58*	7.57
3000	1.84	5.64	4.17	0.83	4.98	5.76
	Type A	Type B			Type C	

*Estimated from ANSI tables by linear interpolation.

Note: Type A—manual or drip lubrication; type B—bath or disk lubrication; type C—oil-stream lubrication.

(*Continued*)

Table 17–20

Rated Horsepower Capacity of Single-Strand Single-Pitch Roller Chain for a 17-Tooth Sprocket (*Continued*)

Sprocket Speed, rev/min		ANSI Chain Number							
		80	100	120	140	160	180	200	240
50	Type A	2.88	5.52	9.33	14.4	20.9	28.9	38.4	61.8
100		5.38	10.3	17.4	26.9	39.1	54.0	71.6	115
150		7.75	14.8	25.1	38.8	56.3	77.7	103	166
200		10.0	19.2	32.5	50.3	72.9	101	134	215
300		14.5	27.7	46.8	72.4	105	145	193	310
400		18.7	35.9	60.6	93.8	136	188	249	359
500	Type B	22.9	43.9	74.1	115	166	204	222	0
600		27.0	51.7	87.3	127	141	155	169	
700		31.0	59.4	89.0	101	112	123	0	
800		35.0	63.0	72.8	82.4	91.7	101		
900		39.9	52.8	61.0	69.1	76.8	84.4		
1000		37.7	45.0	52.1	59.0	65.6	72.1		
1200		28.7	34.3	39.6	44.9	49.9	0		
1400		22.7	27.2	31.5	35.6	0			
1600		18.6	22.3	25.8	0				
1800		15.6	18.7	21.6					
2000		13.3	15.9	0					
2500		9.56	0.40						
3000		7.25	0						
		Type C			Type C′				

Table 17–21

Single-Strand Sprocket Tooth Counts Available from One Supplier*

No.	Available Sprocket Tooth Counts
25	8-30, 32, 34, 35, 36, 40, 42, 45, 48, 54, 60, 64, 65, 70, 72, 76, 80, 84, 90, 95, 96, 102, 112, 120
35	4-45, 48, 52, 54, 60, 64, 65, 68, 70, 72, 76, 80, 84, 90, 95, 96, 102, 112, 120
41	6-60, 64, 65, 68, 70, 72, 76, 80, 84, 90, 95, 96, 102, 112, 120
40	8-60, 64, 65, 68, 70, 72, 76, 80, 84, 90, 95, 96, 102, 112, 120
50	8-60, 64, 65, 68, 70, 72, 76, 80, 84, 90, 95, 96, 102, 112, 120
60	8-60, 62, 63, 64, 65, 66, 67, 68, 70, 72, 76, 80, 84, 90, 95, 96, 102, 112, 120
80	8-60, 64, 65, 68, 70, 72, 76, 78, 80, 84, 90, 95, 96, 102, 112, 120
100	8-60, 64, 65, 67, 68, 70, 72, 74, 76, 80, 84, 90, 95, 96, 102, 112, 120
120	9-45, 46, 48, 50, 52, 54, 55, 57, 60, 64, 65, 67, 68, 70, 72, 76, 80, 84, 90, 96, 102, 112, 120
140	9-28, 30, 31, 32, 33, 34, 35, 36, 37, 39, 40, 42, 43, 45, 48, 54, 60, 64, 65, 68, 70, 72, 76, 80, 84, 96
160	8-30, 32–36, 38, 40, 45, 46, 50, 52, 53, 54, 56, 57, 60, 62, 63, 64, 65, 66, 68, 70, 72, 73, 80, 84, 96
180	13-25, 28, 35, 39, 40, 45, 54, 60
200	9-30, 32, 33, 35, 36, 39, 40, 42, 44, 45, 48, 50, 51, 54, 56, 58, 59, 60, 63, 64, 65, 68, 70, 72
240	9-30, 32, 35, 36, 40, 44, 45, 48, 52, 54, 60

Table 17–22

Tooth Correction Factors, K_1

Number of Teeth on Driving Sprocket	K_1 Pre-extreme Horsepower	K_1 Post-extreme Horsepower
11	0.62	0.52
12	0.69	0.59
13	0.75	0.67
14	0.81	0.75
15	0.87	0.83
16	0.94	0.91
17	1.00	1.00
18	1.06	1.09
19	1.13	1.18
20	1.19	1.28
N	$(N_1/17)^{1.08}$	$(N_1/17)^{1.5}$

Table 17–23

Multiple-Strand Factors, K_2

Number of Strands	K_2
1	1.0
2	1.7
3	2.5
4	3.3
5	3.9
6	4.6
8	6.0